

ARES PdM — system architecture

DATA

NASA C-MAPSS
FD001-FD004 (scripts/fetch_cmapss.py)

Synthetic UAV fleet
20×10 flights (ml/data/uav_synth.py)

TRAIN

Unified training driver — ml/train.py
AdamW · cosine→0 · seed 1337 | LSTM · 1D-CNN · Transformer × {C-MAPSS, UAV} = 6 checkpoints

XAI + EXPORT

Explainability — ml/xai.py
final-layer attention · Captum IG

attention + IG

Bake artifacts — ml/export.py
_batched_predict · 12-asset roster

CONTRACT

Static-artifact contract — public/data/

manifest.json

assets/<id>.json

results.json

SERVE

Next.js dashboard
Armory · Mission · Diagnostics · Lab
(reads static JSON · PROCEDURAL mode)

WS /stream (LIVE)

FastAPI backend :8001 — backend/main.py
synth_stream + inference
(loads UAV Transformer checkpoint)

loads checkpoint

static read